

Memory Hierarchy

- Memory Hierarchy is one of the most required things in Computer Memory as it helps in optimizing the memory available in the computer.

There are multiple levels present in the memory, each one having a different size, different cost, etc.

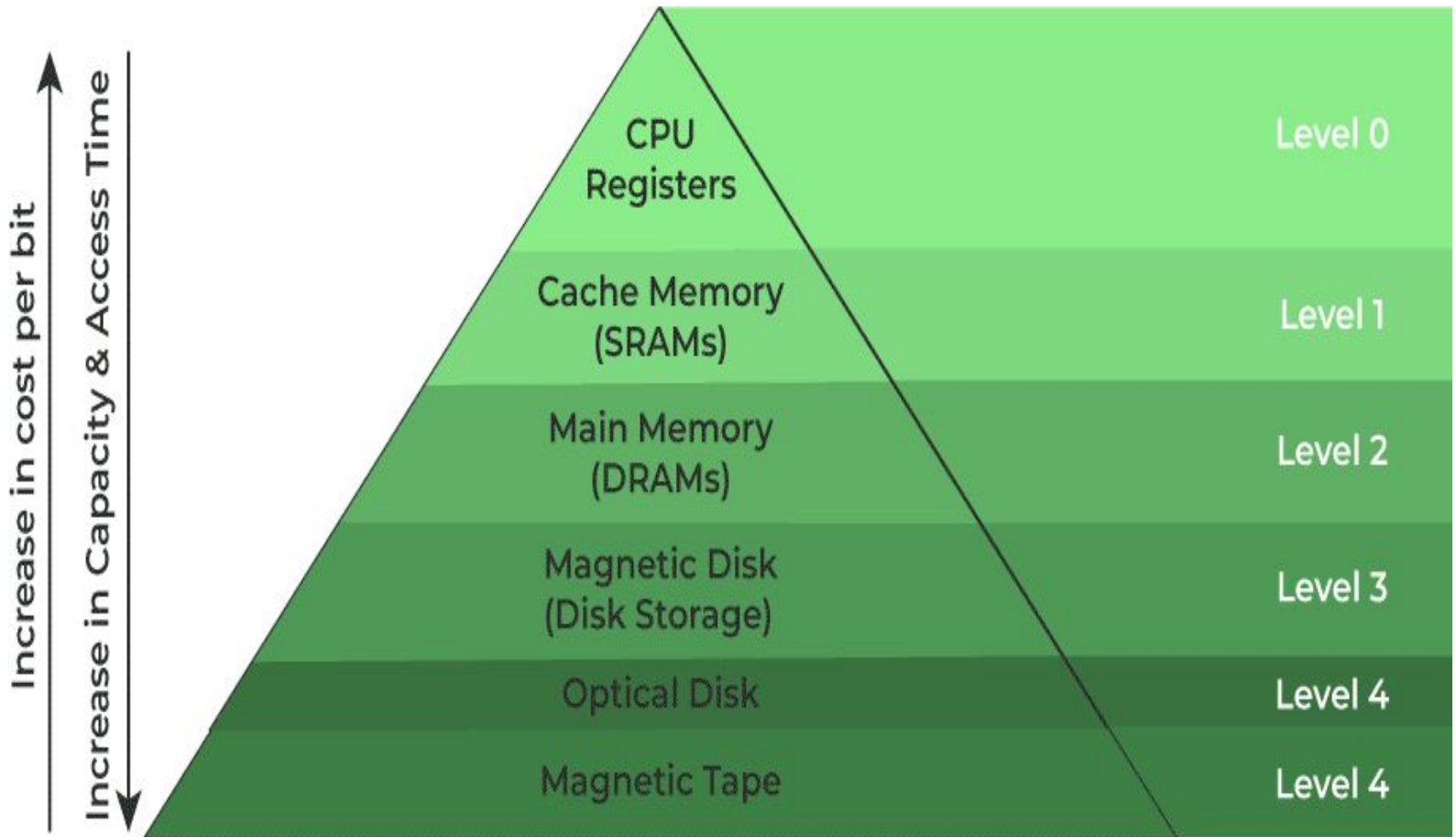
- Some types of memory like cache, and main memory are faster as compared to other types of memory but they are having a little less size and are also costly whereas some memory has a little higher storage value, but they are a little slower.
- Accessing of data is not similar in all types of memory, some have faster access whereas some have slower access.

Types of memory hierarchy

- **External Memory or Secondary Memory:** Comprising of Magnetic Disk, Optical Disk, and Magnetic Tape i.e. peripheral storage devices which are accessible by the processor via an I/O Module.
- **Internal Memory or Primary Memory:** Comprising of Main Memory, Cache Memory & CPU registers. This is directly accessible by the processor.
- The memory technology and storage organization at each level are characterized by five parameters:
 - i. Access time.
 - ii. Memory size.
 - iii. Cost per byte.
 - iv. Transfer bandwidth.
 - v. Unit of transfer.
- ◆ **The unit of transfer in memory hierarchy is the number of data lines that move into and out of a memory module. It can also refer to the number of bits that can be read or written to memory at a time.**

Four level memory hierarchy

Design



Memory Hierarchy Design

Memory Hierarchy

Design

- **1. Registers**
- Registers are small, high-speed memory units located in the CPU. They are used to store the most frequently used data and instructions. Registers have the fastest access time and the smallest storage capacity, typically ranging from 16 to 64 bits.
- **2. Cache Memory**
- Cache memory is a small, fast memory unit located close to the CPU. It stores frequently used data and instructions that have been recently accessed from the main memory. Cache memory is designed to minimize the time it takes to access data by providing the CPU with quick access to frequently used data.
- **3. Main Memory**
- Main memory also known as RAM (Random Access Memory), is the primary memory of a computer system. It has a larger storage capacity than cache memory, but it is slower. Main memory is used to store data and instructions that are currently in use by the CPU.
- **Types of Main Memory**
- **Static RAM:** it stores the binary information in flip flops and information remains valid until power is supplied. It has a faster access time and is used in implementing cache memory.
- **Dynamic RAM:** It stores the binary information as a charge on the capacitor. It requires refreshing circuitry to maintain the charge on the capacitors after a few milliseconds. It contains more memory cells per unit area as compared to SRAM.

4. Secondary Storage

- Secondary storage, such as hard disk drives (HDD) and solid-state drives (SSD) , is a non-volatile memory unit that has a larger storage capacity than main memory. It is used to store data and instructions that are not currently in use by the CPU.
- Secondary storage has the slowest access time and is typically the least expensive type of memory in the memory hierarchy.
- **5. Magnetic Disk**
- Magnetic Disks are simply circular plates that are fabricated with either a metal or a plastic or a magnetized material. The Magnetic disks work at a high speed inside the computer and these are frequently used.
- **6. Magnetic Tape**
- Magnetic Tape is simply a magnetic recording device that is covered with a plastic film. It is generally used for the backup of data. In the case of a magnetic tape, the access time for a computer is a little slower and therefore, it requires some amount of time for accessing the strip.

Advantages of Memory Hierarchy

- It helps in removing some destruction, and managing the memory in a better way.
- It helps in spreading the data all over the computer system.
- It saves the consumer's price and time.

Memory Hierarchy

Information stored in a memory hierarchy ($M_1, M_2, \dots, M_n$) satisfies three important properties:

- **Inclusion Property:** it implies that all information items are originally stored in level M_n . During the processing, subsets of M_n are copied into M_{n-1} . similarly, subsets of M_{n-1} are copied into M_{n-2} , and so on.
- **Coherence Property:** it requires that copies of the same information item at successive memory levels be consistent.
- If a word is modified in the cache, copies of that word must be updated immediately or eventually at all higher levels..
- **Locality of References:** the memory hierarchy was developed based on a program behavior known as locality of references. Memory references are generated by the CPU for either instruction or data access.
- Frequently used information is found in the lower levels

Memory Capacity Planning

- The performance of memory hierarchy is determined by the effective access time.
- The memory capacity planning depends on the hit ratios and access frequencies at successive levels.

Hit Ratio: When an information item is found in M_i it is called as a hit otherwise a miss.

Effective Access time: The performance of a memory hierarchy is determined by **the effective access time (T_{eff})** to any level in the hierarchy. It depends on the hit ratio and access frequencies at successive levels.

$$T_{\text{eff}} = \sum_{i=1}^n f_i \cdot t_i$$

Design of memory hierarchy

- The total cost of memory hierarchy is estimated as follows:

$$C_{\text{total}} = \sum_{i=1}^n c_i \cdot s_i$$

Consider the design of a 3 level memory hierarchy with the following specifications For memory characteristics:

MEMORY LEVEL	ACCESS TIME	CAPACITY	COST/KBYTES
CACHE	t1 = 25ns	s1 = 512kb	c1 =\$0.12
MAIN MEMORY	t2 = unknown	s2 = 32 mb	c2 = \$0.02
DISK ARRAY	t3 = 4ms	s3 = unknown	c3 =\$0.00002

Design memory hierarchy with $t_{\text{eff}}=850\text{ns}$ cache hit ratio of 0.98 and a hit ratio $h_2=0.99$ in main memory. The total cost of memory hierarchy must not exceed \$1500.

*the effective memory access time can be calculated as:

$$h_1 t_1 + (1-h_1)h_2 t_2 + (1-h_1)(1-h_2)h_3 t_3 \leq t_{\text{eff}}$$

Virtual Memory

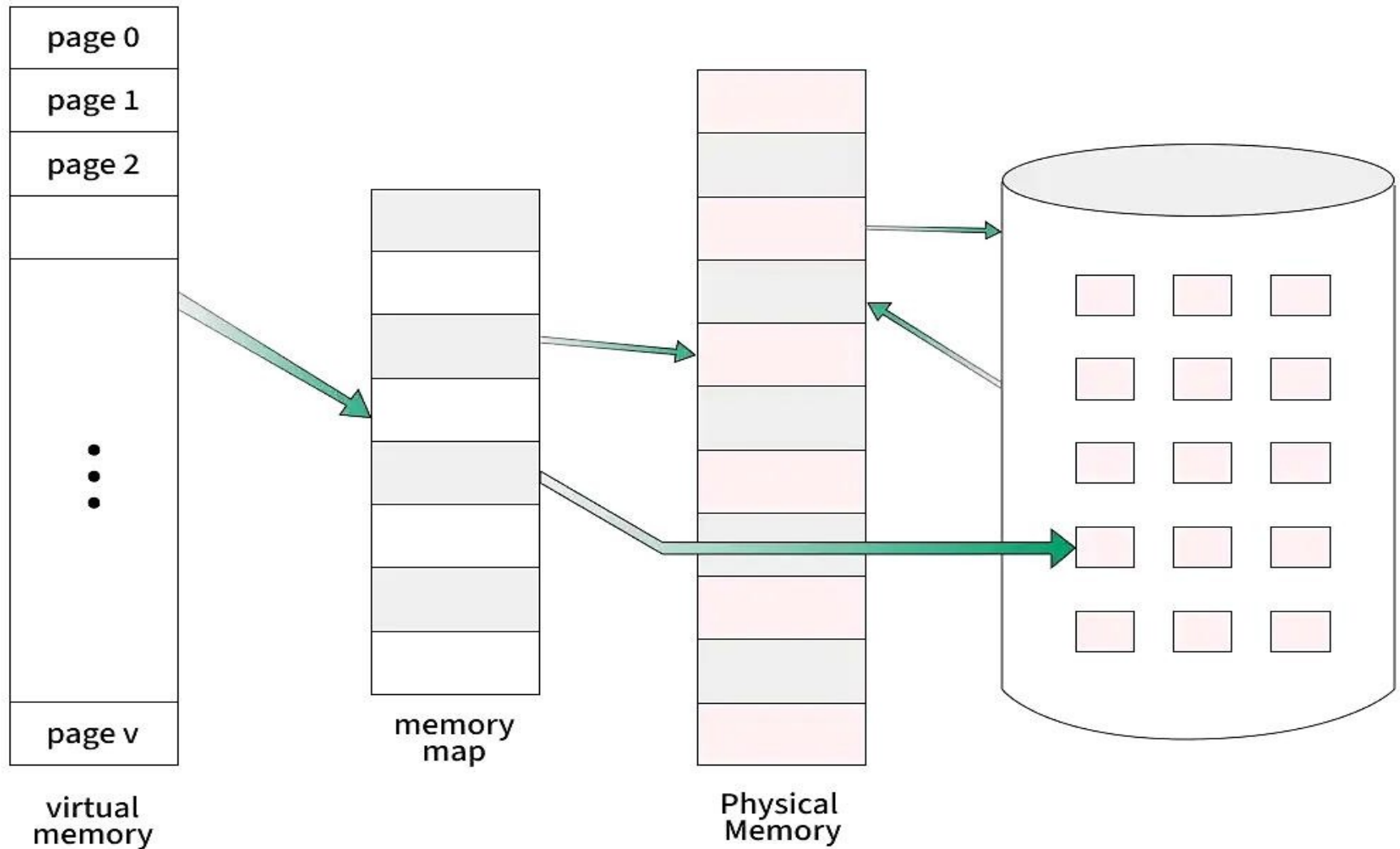
Technology

- Virtual memory is a memory management technique that allows computers to use **secondary memory** as if it were the main memory. It's a common feature of a computer's operating system (OS).
- Virtual memory uses hardware and software to move data between RAM and disk storage.
- This frees up RAM for other programs to use.
- Virtual memory allows computers to run programs that require more memory than the computer's RAM.
- Virtual memory allows computers to multitask and run programs smoothly.
- It's especially useful when a computer needs to run multiple programs at the same time.
- Virtual memory uses a paging file to store information that doesn't fit in RAM.
- The paging file is shared by all processes.
- A virtual address is made up of a page number and an offset within the page.

Limitations

- Virtual memory is slower than RAM.
- Relying too much on virtual memory can cause inconsistencies in the system.

A memory hierarchy, consisting of a computer system's memory and a disk, enables a process to operate with only some portions of its address space in RAM to allow more processes to be in memory.



Types of Virtual

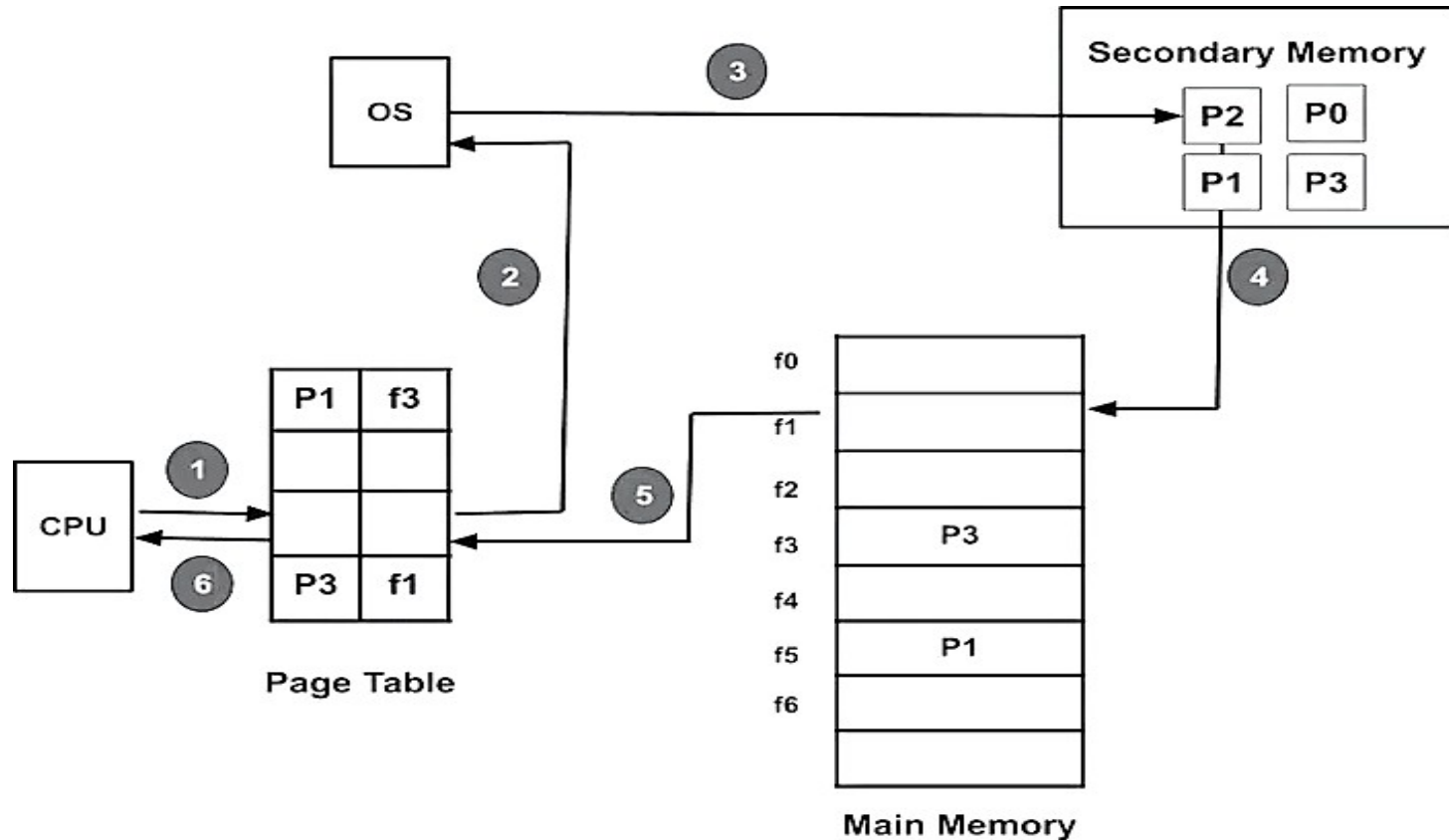
In a computer, virtual memory is managed by the Memory Management Unit (MMU), which is often built into the CPU. The CPU generates virtual addresses that the MMU translates into physical addresses.

There are two main types of virtual memory:

Paging

- Paging divides memory into small fixed-size blocks called pages.
- When the computer runs out of RAM it uses pages to use as a physical memory.
- pages that aren't currently in use are moved to the hard drive, into an area called a swap file.
- The swap file acts as an extension of RAM. When a page is needed again, it is swapped back into RAM, a process known as page swapping. This ensures that the operating system (OS) and applications have enough memory to run.

Demand Paging: The process of loading the page into memory on demand (whenever a page fault occurs) is known as demand paging.



Page fault service time: The time taken to service the page fault is called page fault service time. The page fault service time includes the time taken to perform all the above six steps.

- Let Main memory access time is: m
- Page fault service time is: s
- Page fault rate is : p
- Then, Effective memory access time = $(p*s) + (1-p)*m$

The other type of memory is segmentation:

- Segmentation divides virtual memory into segments of different sizes.
- Segments that aren't currently needed can be moved to the hard drive.
- The system uses a segment table to keep track of each segment's status, including whether it's in memory, if it's been modified, and its physical address.
- Management of segmented memory system is more complex due to non uniform segment size.

Virtual shared memory and shared virtual memory (SVM)

are programming models that allow processors to share memory. They are used in distributed-memory machines and OpenCL 2.0, respectively.

Virtual shared memory

- A programming model that allows processors to be programmed as if they share memory
- A software layer handles communication between processors
- Accessing shared variables requires communication over a network, which is slower than accessing local variables

- **Shared virtual memory (SVM)**
- A feature of OpenCL 2.0 that allows the host and device to share memory
- Allows the use of pointers and complex data structures system-wide
- Eliminates the need for explicit transfers between the host and device
- Has two types: fine-grained and coarse-grained.
- **fine-grained** access allows data sharing at the byte level with automatic consistency.
- **coarse-grained** sharing synchronizes only at explicit points (like kernel launches or map/unmap).
- Coarse-grained is the basic requirement, whereas fine-grained is an optional, more advanced feature for better performance with